PHYS 217 Electricity and Magnetism I

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1 Vector Analysis

1.1 Vector Algebra

Definition 1.1. A **vector** is a quantity that has both direction and magnitude.

Definition 1.2. A **scalar** is a quantity that only has magnitude.

1.1.1 Vector Operations

There are four vector operations:

Addition:

- $\mathbf{A} + \mathbf{B}$ is the vector from the start of $\mathbf{A}$ to the end of $\mathbf{B}$
- Commutative: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
- Associative: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$
- Subtracting a vector is just adding its opposite: $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$

Multiplication by Scalar:

- $a\mathbf{A}$ is a vector with the same direction as $\mathbf{A}$ but a times the magnitude
- Distributive: $a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$

Dot Product (Inner Product or Scalar Product):

- $\mathbf{A} \cdot \mathbf{B} \equiv AB\cos\theta$
- Results in scalar
- Commutative: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- Distributive: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
- Geometrically, $\mathbf{A} \cdot \mathbf{B}$ is the product of A and the projection of $\mathbf{A}$ along $\mathbf{B}$

Cross Product of Two Vectors:

- $\mathbf{A} \times \mathbf{B} \equiv AB\sin\theta\hat{\mathbf{n}}$ where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$. The direction is determined by the **right hand rule**, where you curl your right hand from $\mathbf{A}$ to $\mathbf{B}$ and take the direction of your thumb
- Distributive: $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C})$
- Not commutative: $\mathbf{B} \times \mathbf{A} = -(\mathbf{A} \times \mathbf{B})$
- Result is a **pseudovector** and behaves like a vector under rigid transformations like rotations, but not under certain nonrigid transformations. Notably, $-\mathbf{A} \times -\mathbf{B} = \mathbf{A} \times \mathbf{B}$
- Geometrically, $|\mathbf{A} \times \mathbf{B}|$ is the area of the parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$

When working in Cartesian coordinates, vectors can be expressed in terms of **basis vectors**:

$$\mathbf{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}$$

with A_x, A_y, and A_z being the **components** of $\mathbf{A}$.

Vector operations can be done with the components, but the reader should already know how to do this.

There are two types of triple vector products:

Scalar Triple Product:

- $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$
- Volume of parallelepiped generated by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$

Vector Triple Product:

- **BAC-CAB rule**: $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$

1.1.2 Nomenclature

Definition 1.3. The **position vector** of a point is the vector to that point from the origin:

$$\mathbf{r} \equiv x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

Definition 1.4. The **infinitesimal displacement vector** from (x, y, z) to $(x + dx, y + dy, z + dz)$ is:

$$d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$$

Definition 1.5. A **separation vector** is the vector from a **source point** to a **field point**:

$$\boldsymbol{\imath} \equiv \mathbf{r} - \mathbf{r}'$$

1.1.3 Vector Transformations

Coordination transformation from rotation about an arbitrary axis can be represented by the use of a **rotation matrix**:

$$\begin{pmatrix} \bar{A}_x \\ \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{yx} & R_{yy} & R_{yz} \\ R_{zx} & R_{zy} & R_{zz} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}$$

or more concisely:

$$\bar{A}_i = \sum_{j=1}^{3} R_{ij} A_j$$

Definition 1.6. A transformation is **unitary** if it does not change the magnitude of the object on which it operates.

If rotation matrix $\overleftrightarrow{R}$ is unitary, then:

$$\sum_{i}^{3} R_{ij} R_{ik} = \delta_{jk}$$

where:

$$\delta_{jk} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

is the Kronecker delta.

Definition 1.7. **Parity** is a property related to how a system is affected by coordinate inversions. If a system stays the same after a parity transformation, it is said to have even parity. If it is inverted, it is said to have odd parity.

1.1.4 Dyadics and Tensors

Definition 1.8. A **tensor** is something that transforms like a tensor.

Definition 1.9. A **dyadic product** or outer product is a type of multiplication that generates a second-order tensor called a **dyadic** from two vectors:

$$\overleftrightarrow{\mathbf{C}} = \mathbf{AB}$$

Dyadics have nine independent components and in Cartesian coordinates can be decomposed into **dyads**:

$$\overleftrightarrow{\mathbf{C}} = A_x B_x \hat{\mathbf{x}}^2 + +A_x B_y \hat{\mathbf{x}}\hat{\mathbf{y}} + A_x B_z \hat{\mathbf{x}}\hat{\mathbf{z}} + A_y B_x \hat{\mathbf{y}}\hat{\mathbf{x}} + A_y B_y \hat{\mathbf{y}}^2 + A_y B_z \hat{\mathbf{y}}\hat{\mathbf{z}} + A_z B_x \hat{\mathbf{z}}\hat{\mathbf{x}} + A_z B_y \hat{\mathbf{z}}\hat{\mathbf{y}} + A_z B_z \hat{\mathbf{z}}^2$$

Thus, dyadics can be represented in matrix form:

$$\overleftrightarrow{\mathbf{C}} = \begin{bmatrix} A_xB_x & A_xB_y & A_xB_z \\ A_yB_x & A_yB_y & A_yB_z \\ A_zB_x & A_zB_y & A_zB_z \end{bmatrix} = \begin{bmatrix} A_x & A_y & A_z \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \mathbf{A}\mathbf{B}^T$$

The transpose symbol is usually omitted in E&M.
A dyadic $\overleftrightarrow{C}$ has two invariants, **spur** (called cusp in lecture) $|\overleftrightarrow{\mathbf{C}}|$ and **rotation vector** $\langle\overleftrightarrow{\mathbf{C}}\rangle$:

$$|\overleftrightarrow{\mathbf{C}}| = \sum_i^3 C_{ii}$$
$$\langle\overleftrightarrow{\mathbf{C}}\rangle = \hat{\mathbf{x}}(C_{yz} - C_{zy}) + \hat{\mathbf{y}}(C_{zx} - C_{xz}) + \hat{\mathbf{z}}(C_{xy} - C_{yx})$$

Dyadics commute in addition and scalar multiplication but not in inner products:

$$\overleftrightarrow{\mathbf{C}} + \overleftrightarrow{\mathbf{D}} = \overleftrightarrow{\mathbf{D}} + \overleftrightarrow{\mathbf{C}}$$
$$a\overleftrightarrow{\mathbf{C}} = \overleftrightarrow{\mathbf{C}}a$$
$$\mathbf{A} \cdot \overleftrightarrow{\mathbf{C}} \neq \overleftrightarrow{\mathbf{C}} \cdot \mathbf{A}$$

Definition 1.10. Every dyadic $\overleftrightarrow{\mathbf{C}}$ has a **conjugate** $\overleftrightarrow{\mathbf{C}}^*$ such that:

$$\mathbf{A} \cdot \overleftrightarrow{\mathbf{C}} = \overleftrightarrow{\mathbf{C}}^* \cdot \mathbf{A}$$

In three dimensional Cartesian coordinates, the conjugate of a dyadic is simply its transpose.

1.2 Differential Calculus

Definition 1.11. The **gradient** of a three dimensional scalar-valued function $f(x, y, z)$ is a vector field that points in the direction of maximum increase of f and has a magnitude equal to the rate of increase:

$$\nabla f = \frac{\partial f}{\partial x}\hat{\mathbf{x}} + \frac{\partial f}{\partial y}\hat{\mathbf{y}} + \frac{\partial f}{\partial z}\hat{\mathbf{z}}$$

Definition 1.12. The **divergence** of a three dimensional vector-valued function $f(x, y, z) = f_x\hat{\mathbf{x}} + f_y\hat{\mathbf{y}} + f_z\hat{\mathbf{z}}$ is a scalar that represents how much the vector field diverges or converges, analogous to sources and sinks in fluid flow:

$$\nabla \cdot f = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}$$

Definition 1.13. The **curl** of a three dimensional vector-valued function $f(x, y, z) = f_x\hat{\mathbf{x}} + f_y\hat{\mathbf{y}} + f_z\hat{\mathbf{z}}$ is a vector field with magnitude representing the infinitesimal circulation, analogous to rotation in a fluid, and direction perpendicular to that circulation:

$$\nabla \times f = (\frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z})\hat{\mathbf{x}} + (\frac{\partial f_x}{\partial z} - \frac{\partial f_z}{\partial x})\hat{\mathbf{y}} + (\frac{\partial f_y}{\partial x} - \frac{\partial f_x}{\partial y})\hat{\mathbf{z}}$$

Gradient, divergence, and curl can all be found by treating the vector operator ∇ **del** as a vector:

$$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}$$

Gradient, divergence, and curl behave like ordinary derivatives under summation and multiplication by scalars. Since gradient, divergence, and curl can be multiplied by scalars and vectors in multiple

ways, there are six product rules:

$$\begin{aligned}
\nabla(fg) &= f\nabla g + g\nabla f \\
\nabla(\mathbf{A}\cdot\mathbf{B}) &= \mathbf{A}\times(\nabla\times\mathbf{B}) + \mathbf{B}\times(\nabla\times\mathbf{A}) + (\mathbf{A}\cdot\nabla)\mathbf{B} + (\mathbf{B}\cdot\nabla)\mathbf{A} \\
\nabla\cdot(f\mathbf{A}) &= f(\nabla\cdot\mathbf{A}) + \mathbf{A}\cdot(\nabla f) \\
\nabla\cdot(\mathbf{A}\times\mathbf{B}) &= \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B}) \\
\nabla\times(f\mathbf{A}) &= f(\nabla\times\mathbf{A}) - \mathbf{A}\times(\nabla f) \\
\nabla\times(\mathbf{A}\times\mathbf{B}) &= (\mathbf{B}\cdot\nabla)\mathbf{A} - (\mathbf{A}\cdot\nabla)\mathbf{B} + \mathbf{A}(\nabla\cdot\mathbf{B}) - \mathbf{B}(\nabla\cdot\mathbf{A})
\end{aligned}$$

Since you can combine the three derivatives in various ways, there are five species of second derivatives, but all but two are either degenerate or are composed of the others:

$$\nabla\cdot(\nabla f)$$
$$\nabla(\nabla\cdot f)$$

Definition 1.14. The **Laplacian** of three dimensional scalar-valued function f is:

$$\nabla^2 f = \nabla\cdot(\nabla f)$$

1.3 Integral Calculus

Definition 1.15. A **line integral** of a vector-valued function is the integral of the dot product of the function and infinitesimal displacement along a curve:

$$\int_a^b \mathbf{v}\cdot d\mathbf{l}$$

Definition 1.16. A **surface integral** of a vector-valued function is the integral of the dot product of the function and infinitesimal area over a surface:

$$\int_S \mathbf{v}\cdot d\mathbf{a}$$

Definition 1.17. A **volume integral** of a scalar-valued function is the integral of the function over a volume:

$$\int_v T d\tau$$

Since there are three types of derivatives in three dimensions, there are three versions of the fundamental theorem of calculus.

Theorem 1.1. Gradient Theorem

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla T)\cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})$$

Proof. left as an exercise for the reader (will do tonight) □

Theorem 1.2. Divergence Theorem or Gauss's Theorem

$$\int_V (\nabla\cdot\mathbf{v}) d\tau = \oint_{\partial V} \mathbf{v}\cdot d\mathbf{a}$$

Proof. left as an exercise for the reader (will finish tonight) □

Theorem 1.3. Stoke's Theorem

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_{\partial S} \mathbf{v} \cdot d\mathbf{l}$$

Proof. left as an exercise for the reader (will finish tonight) □

1.4 Curvilinear Coordinates

Definition 1.18. **Spherical coordinates** are a coordinate system that uses the distance from the origin r, the angle down from the z-axis called the **polar angle** θ, and the angle around from the x-axis called the **azimuthal angle** ϕ.

The following are true for spherical coordinates:

- Conversion to Cartesian coordinates: $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$
- Unit vectors in terms of Cartesian coordinates:

$$\begin{aligned}\hat{\mathbf{r}} &= \sin\theta\cos\phi\hat{\mathbf{x}} + \sin\theta\sin\phi\hat{\mathbf{y}} + \cos\theta\hat{\mathbf{z}}\\ \hat{\theta} &= \cos\theta\cos\phi\hat{\mathbf{x}} + \cos\theta\sin\phi\hat{\mathbf{y}} - \sin\theta\hat{\mathbf{z}}\\ \hat{\phi} &= -\sin\phi\hat{\mathbf{x}} + \cos\phi\hat{\mathbf{y}}\end{aligned}$$

- Infinitesimal displacement: $d\hat{\mathbf{l}} = dr\hat{\mathbf{r}} + rd\theta\hat{\theta} + r\sin\theta d\phi\hat{\phi}$
- Infinitesimal volume: $d\tau = r^2\sin\theta dr d\theta d\phi$
- Gradient: $\nabla T = \frac{\partial T}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial T}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial T}{\partial\phi}\hat{\phi}$
- Divergence:$\nabla\cdot\mathbf{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta v_\theta) + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial\phi}$
- Curl: $\nabla\times\mathbf{v} = \frac{1}{r\sin\theta}[\frac{\partial}{\partial\theta}(\sin\theta v_\phi) - \frac{\partial v_\theta}{\partial\phi}]\hat{\mathbf{r}} + \frac{1}{r}[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial\phi} - \frac{\partial}{\partial r}(rv_\phi)]\hat{\theta} + \frac{1}{r}[\frac{\partial}{\partial r}(rv_\theta) - \frac{\partial v_r}{\partial\theta}]\hat{\phi}$
- Laplacian: $\nabla^2 T = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial T}{\partial r}) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial T}{\partial\theta}) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 T}{\partial\phi^2}$

Definition 1.19. **Cylindrical coordinates** are a coordinate system that uses distance from the z-axis s, the angle around from the x-axis ϕ, and the distance up the z-axis z.

The following are true for cylindrical coordinates:

- Conversion to Cartesian coordinates: $x = s\cos\phi$ $y = s\sin\phi$, $z = z$
- Unit vectors in terms of Cartesian coordinates:

$$\begin{aligned}\hat{\mathbf{s}} &= \cos\phi\hat{\mathbf{x}} + \sin\phi\hat{\mathbf{y}}\\ \hat{\phi} &= -\sin\phi\hat{\mathbf{x}} + \cos\phi\hat{\mathbf{y}}\\ \hat{\mathbf{z}} &= \hat{\mathbf{z}}\end{aligned}$$

- Infinitesimal displacement: $d\mathbf{l} = ds\hat{\mathbf{s}} + sd\phi\hat{\phi} + dz\hat{\mathbf{z}}$
- Infinitesimal volume: $d\tau = sdsd\phi dz$
- Gradient: $\nabla T = \frac{\partial T}{\partial s}\hat{\mathbf{s}} + \frac{1}{s}\frac{\partial T}{\partial\phi}\hat{\phi} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}$
- Divergence: $\nabla\cdot\mathbf{v} = \frac{1}{s}\frac{\partial}{\partial s}(sv_s) + \frac{1}{s}\frac{\partial v_\phi}{\partial\phi} + \frac{\partial v_z}{\partial z}$
- Curl: $\nabla\times\mathbf{v} = (\frac{1}{s}\frac{\partial v_z}{\partial\phi} - \frac{\partial v_\phi}{\partial z})\hat{\mathbf{s}} + (\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s})\hat{\phi} + \frac{1}{s}[\frac{\partial}{\partial s}(sv_\phi) - \frac{\partial v_s}{\partial\phi}]\hat{\mathbf{z}}$
- Laplacian: $\nabla^2 T = \frac{1}{s}\frac{\partial}{\partial s}(s\frac{\partial T}{\partial s}) + \frac{1}{s^2}\frac{\partial^2 T}{\partial\phi^2} + \frac{\partial^2 T}{\partial z^2}$

1.5 Dirac Delta Function

Definition 1.20. The **Dirac delta function** is a **generalized function** or **distribution** with a value of 0 everywhere except 0, where it equals infinity:

$$\delta(x) = \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases}$$

The delta function has the integral:

$$\int_{-\infty}^{\infty} \delta(x)dx = 1$$

The Dirac delta function is useful because it allows us to pick out a specific value of a function in an integral:

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = f(a)$$

The Dirac delta function behaves similarly in three dimensions and can be written as:

$$\delta^3(\boldsymbol{\imath}) = \frac{1}{4\pi}\nabla \cdot (\frac{\hat{\boldsymbol{\imath}}}{\boldsymbol{\imath}^2})$$

1.6 Theory of Vector Fields

Theorem 1.4. **Helmholtz Theorem**

If the divergence $D(\mathbf{r})$ and curl $C(\mathbf{r})$ of a vector function $\mathbf{F}(\mathbf{r})$ are specified, and if both go to zero faster than $\frac{1}{r^2}$ as $r \to \infty$, and if $\mathbf{F}(\mathbf{r})$ goes to zero as $r \to \infty$, then $\mathbf{F}$ can uniquely be decomposed into an irrotational component $-\nabla U$ and solenoidal component $\nabla \times \mathbf{W}$:

$$\mathbf{F} = -\nabla U + \nabla \times \mathbf{W}$$

Theorem 1.5. **Theorem 1. Irrotational Fields**

The following conditions are equivalent:

- $\nabla \times \mathbf{F} = 0$ everywhere
- $\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l}$ is independent of path for any given end points
- $\oint \mathbf{F} \cdot d\mathbf{l} = 0$ for any closed loop
- $\mathbf{F}$ is the gradient of some scalar function V called the **scalar potential**: $\mathbf{F} = -\nabla V$

Theorem 1.6. **Theorem 2. Solenoidal Fields**

The following conditions are equivalent:

- $\nabla \cdot \mathbf{F} = 0$ everywhere
- $\int \mathbf{F} \cdot d\mathbf{a}$ is independent of surface for a given boundary line
- $\oint \mathbf{F} \cdot d\mathbf{a} = 0$ for any closed surface
- $\mathbf{F}$ is the curl of some vector function $\mathbf{A}$ called the **vector potential**: $\mathbf{F} = \nabla \times \mathbf{A}$

2 Electrostatics

2.1 The Electric Field

Definition 2.1. The **principle of superposition** states that the interaction between any two charges is completely unaffected by the presence of other charges.

Law 2.1. Coulomb's Law
The force on charge Q from charge q with separation vector $\boldsymbol{\imath}$ between them is:

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0}\frac{qQ}{\boldsymbol{\imath}^2}\hat{\boldsymbol{\imath}}$$

where ϵ_0 is the **permittivity of free space**.

Definition 2.2. An **electric field E** is the total contribution from a collection of charges or a continuous charge distribution, with the resultant force on test charge Q being:

$$\mathbf{F} = Q\mathbf{E}$$

The electric field of a volume charge with charge density ρ is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{r}')}{\boldsymbol{\imath}^2}\hat{\boldsymbol{\imath}}d\tau'$$

2.2 Divergence and Curl of Electrostatic Fields

We notice that the **electric flux** $\int \mathbf{E}\cdot d\mathbf{a}$ through a closed surface only depends on the divergence of the electric field within that surface, since any fields originating from outside pass through two sides of the surface, making the net flux zero.

Law 2.2. Gauss's Law

Integral form:

$$\oint \mathbf{E}\cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0}$$

Differential form:

$$\nabla\cdot\mathbf{E} = \frac{\rho}{\epsilon_0}$$

We can also derive this by taking the divergence of the formula for the electric field from a charge distribution:

$$\begin{aligned}\nabla\cdot\mathbf{E} &= \frac{1}{4\pi\epsilon_0}\int \nabla\cdot\left(\frac{\hat{\boldsymbol{\imath}}}{\boldsymbol{\imath}^2}\right)\rho(\mathbf{r}')d\tau' \\ &= \frac{1}{4\pi\epsilon_0}\int 4\pi\delta^3(\mathbf{r}-r')\rho(\mathbf{r}')d\tau' \\ &= \frac{\rho(\mathbf{r})}{\epsilon_0}\end{aligned}$$

When a situation has symmetry, Gauss's law can be used to easily find the electric field. Additionally, due to the principle of superposition, some situations can be split into multiple problems that have symmetry and are thus simple to solve.

2.3 Electric Potential

Definition 2.3. The **electric potential** is defined as:

$$V(\mathbf{r}) \equiv -\int_{\mathscr{O}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}$$

where $\mathscr{O}$ is a reference point where the potential is 0 (typically at infinity). This can be thought of as the work required to move a unit charge from infinity to $\mathbf{r}$.

The differential form of this is:

$$\mathbf{E} = -\nabla V$$

Electric potential is useful because it is a scalar function, not a vector function (though the electric field still only has one independent function because it has zero curl). Like the electric field, it obeys the principle of superposition.

Definition 2.4. **Poisson's Equation for Electrostatics**

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}$$

Definition 2.5. **Laplace's Equation**

$$\nabla^2 V = 0$$

The potential of a charge distribution is:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(r')}{\imath} d\tau'$$

The electric field experiences discontinuities across surface charges (easily verified using Gauss's law) but the tangential component is continuous across surface charges (also easily verified with Gauss's law), so:

$$E^{\perp}_{above} - E^{\perp}_{below} = \frac{\sigma}{\epsilon_0}\hat{\mathbf{n}}$$

This can also be written in terms of the electric potential:

$$\frac{\partial V_{above}}{\partial n} - \frac{\partial V_{below}}{\partial n} = -\frac{\sigma}{\epsilon_0}$$

2.4 Work and Energy in Electrostatics

Starting with a single point charge, we can incrementally move point charges from infinity to different positions, with the total amount of work done being the energy of the configuration:

$$W = \frac{1}{2}\sum_{i=1}^{n} q_i V(\mathbf{r}_i)$$

We can determine a version of this for continuous charge distributions:

$$\begin{aligned}
W &= \frac{1}{2}\int \rho V d\tau \\
&= \frac{\epsilon_0}{2}\int (\nabla \cdot \mathbf{E}) V d\tau \quad \text{using Gauss's law} \\
&= \frac{\epsilon_0}{2}[-\int_{\mathscr{V}} \mathbf{E} \cdot (\nabla V) d\tau + \oint_{\mathscr{S}} V\mathbf{E} \cdot d\mathbf{a}] \quad \text{using integration by parts} \\
&= \frac{\epsilon_0}{2}(\int_{\mathscr{V}} E^2 d\tau + \oint_{\mathscr{S}} V\mathbf{E} \cdot d\mathbf{a}) \\
&= \frac{\epsilon_0}{2}\int E^2 d\tau \quad \text{if E vanishes at infinity}
\end{aligned}$$

2.5 Conductors

In **insulators**, electrons are attached to particular atoms, but in metallic **conductors**, electrons can move around freely. This leads to the following properties:

- $\mathbf{E} = 0$ inside a conductor (because otherwise charges would flow until this is the case). In the presence of external electric fields, **induced charges** are created that move to produce electric fields that counteract the external field
- $\rho = 0$ inside a conductor
- All net charge lies on the surface
- Conductors are equipotentials
- $\mathbf{E}$ is perpendicular to the surface just outside a conductor

The field immediately outside a conductor is:

$$\mathbf{E} = \frac{\sigma}{\epsilon_0}\hat{\mathbf{n}}$$

And equivalently:

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$$

The force on the surface of an electric conductor is:

$$\mathbf{F} = \sigma \mathbf{E}_{average} = \frac{1}{2}(\mathbf{E}_{above} + \mathbf{E}_{below}) = \frac{1}{2\epsilon_0}\sigma^2 \hat{\mathbf{n}}$$

And the **electrostatic pressure** pointing outward is thus:

$$P = \frac{\epsilon_0}{2}E^2$$

If we have two conductors and put a charge of $+Q$ on one and $-Q$ on the other, the potential difference between the capacitors is proportional to Q, and the ratio is the **capacitance**:

$$C \equiv \frac{Q}{V}$$

The energy stored by the capacitor is the total work required to move the positive charge to the positive conductor and negative charge to the negative conductor, or:

$$W = \int_0^Q (\frac{q}{C})dq = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$$

3 Potentials

3.1 Laplace's Equation

It is often difficult to find the electric field or potential analytically through integration, and thus in many cases it is better to use the Poisson equation. In regions with zero charge, this reduces to the Laplace equation, solutions of which are called **harmonic functions**.

Harmonic functions have several useful properties:

- The value a harmonic function V at point $\mathbf{r}$ is the average value of V over a spherical surface of radius R centered at $\mathbf{r}$:
$$V(\mathbf{r}) = \frac{1}{4\pi R^2} \oint_{sphere} V da$$
- V cannot have any local minima or maxima; extrema must occur at boundaries

Theorem 3.1. Earnshaw's Theorem
A charged particle cannot be held in stable equilibrium by electrostatic forces alone.

Proof. Imagine a cube with point charges at the vertices. The electric potential in the interior of the cube satisfies the Laplace equation because there is no charge there, and thus cannot have a minimum inside the cube, meaning there cannot be a stable equilibrium. □

For the Laplace equation to be useful, we need to know whether solutions exist and whether they are unique (and thus we can use whatever method we prefer and not worry about determining if we found the right solution). The existence of solutions is difficult to prove and left to mathematicians, but we can relatively easily prove that solutions are unique.

Theorem 3.2. Uniqueness Theorem
The Poisson equation, when subjected to boundary conditions on the solution, its first derivative, or combinations of these, yields unique solutions for $V(\mathbf{r})$.

Proof. Let V be continuous, as well as its derivatives through second order in region $\mathscr{V}$ bounded by $\partial\mathscr{V}$. Suppose:

$$\nabla^2 V = f$$
$$p\frac{dV}{dn} + hV = g$$

where f is a function in $\mathscr{V}$ and g is a function on $\partial\mathscr{V}$. (Note: the second equation is the **Robin boundary condition**, which is a generalization of Dirichlet and Neumann boundary conditions).

If V_1 and V_2 are both solutions, then by superposition, $U \equiv V_1 - V_2$ is a solution to the Laplace equation:

$$\nabla^2 U = \nabla^2 V_1 - \nabla^2 V_2 = 0$$
$$p\frac{dU}{dn} + hV = 0 - 0 = 0$$

Let us consider the function $U\nabla U$ with divergence $\nabla \cdot (U\nabla U) = U\nabla^2 U + (\nabla U)^2$. The integral of this over $\mathscr{V}$ is:

$$\int_{\mathscr{V}} \nabla \cdot (U\nabla U) d\tau = \int_{\mathscr{V}} \underbrace{U\nabla^2 U}_{\text{U is harmonic}}^{0} + (\nabla U)^2 d\tau = \int_{\mathscr{V}} (\nabla U)^2 d\tau = \oint_{\partial\mathscr{V}} U\nabla U \cdot d\mathbf{a} = \oint_{\partial\mathscr{V}} U\frac{dU}{dn} da$$

There are three non-trivial cases of the boundary conditions:

1. $p = 0$ and thus only V is defined on the boundary
From $p\frac{dU}{dn} + hU = 0$, this means that $U = 0$ everywhere on $\partial\mathscr{V}$. Then, since $(\nabla U)^2$ can't be negative, $\nabla U = 0$, so $U = 0$ everywhere on $\mathscr{V}$. Therefore $V_1 = V_2$ and the solution is unique.

2. $h = 0$ and thus only $\frac{dV}{dn}$ is defined on the boundary
From $p\frac{dU}{dn} + hU = 0$, this means that $\frac{dU}{dn} = 0$ everywhere on $\partial\mathscr{V}$. Then $\nabla U = 0$ again, so U is constant on $\mathscr{V}$. Therefore V_1 and V_2 only differ by an additive constant (reference point).

3. $p\frac{dU}{dn} + hU = 0$ with mixed boundary conditions (both V and $\frac{dV}{dn}$ defined)
From $p\frac{dU}{dn} + hU = 0$, we find that $\frac{dU}{dn} = -\frac{h}{p}U$ and thus $\oint_{\partial\mathscr{V}} U\frac{dU}{dn}da = -\frac{h}{p}\oint_{\partial\mathscr{V}} U^2 da \leq 0$. However, we know that $\oint_{\partial\mathscr{V}} U\nabla U \cdot d\mathbf{a} = \oint_{\partial\mathscr{V}} U\frac{dU}{dn}da = \int_{\mathscr{V}}(\nabla U)^2 d\tau \geq 0$, and thus $\nabla U = 0$ and U is constant, meaning that V_1 and V_2 differ only by an additive constant.

□

A useful corollary of the uniqueness theorem is that if we have a volume $\mathscr{V}$ surrounded by conductors and containing a specified charge density ρ, then the electric field is uniquely determined if the total charge on each conductor is given.

3.2 Method of Images

In many situations, you can use a technique reminiscent of geometrical optics called the **method of images**. Conductors (or other equipotential surfaces) are treated as mirrors, and virtual charges are added at the locations of images of the real charges, and as long as the charges are arranged in such a way that the boundary conditions, including constant potential on the conductors, holds, the solutions are valid. These images must be real and the solution only holds in the region outside of the conductor.

Example. Suppose a point charge q is placed a distance d above an infinite grounded conducting plane. What is the potential in the region above the plane?

Answer. We can treat the conducting plane as a mirror. As we know from basic optics, the image of the point charge is a distance d below the plane, at the same x and y coordinates as the original charge. The potential in this situation is:

$$V(x,y,z) = \frac{1}{4\pi\epsilon_0}\left[\frac{q}{\sqrt{x^2+y^2+(z-d^2)}} - \frac{q}{\sqrt{x^2+y^2+(z+d)^2}}\right]$$

This reduces to zero at $z = 0$, matching the boundary condition for the infinite grounded conducting plane, meaning that this solution is valid, and by the uniqueness of solutions to the Poisson equation, is the only solution for this scenario. Thus, we can use it for $z \geq 0$. ⊛

3.3 Separation of Variables

One of the simplest methods of solving the Laplace equation is **separation of variables**, where V is decomposed into a product of functions in each dimension. This can be used when the potential or charge density are specified on the boundaries of some region where we want to find the potential of the interior. It is easiest to derive and elucidate this method with several examples:

3.3.1 Cartesian Coordinates

Example. Two infinite grounded metal plates lie parallel to the xz plane, one at $y = 0$ and the other at $y = a$. The left end, at $x = 0$, is closed off with an infinite strip insulated from the two plates, and maintained at a specific potential $V_0(y)$. Find the potential inside this "slot".

Answer. Since this configuration is independent of z, this is effectively a two dimensional problem, with the Laplace equation:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

And boundary conditions:

$$\begin{aligned} V(x,0) &= 0 \\ V(x,a) &= 0 \\ V(0,y) &= V_0(y) \\ V &\to 0 \text{ as } x \to \infty \end{aligned}$$

Suppose that the solution is the product of functions in x and y:

$$V(x,y) = X(x)Y(y)$$

Substituting this into the Laplace equation, we get:

$$\begin{aligned} Y\frac{\partial^2 X}{\partial x^2} + X\frac{\partial^2 Y}{\partial Y^2} &= 0 \\ \to \frac{1}{X}\frac{\partial^2 X}{\partial x^2} + \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} &= 0 \end{aligned}$$

This can only be true if $\frac{1}{X}\frac{\partial^2 X}{\partial x^2}$ and $\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2}$ both equal the same constant, so:

$$\begin{aligned} \frac{\partial^2 X}{\partial x^2} &= k^2 X \\ \frac{\partial^2 Y}{\partial y^2} &= -k^2 Y \end{aligned}$$

We have thus reduced our situation to two ODEs. We know that the general solutions are of the form:

$$\begin{aligned} X(x) &= Ae^{kx} + Be^{-kx} \\ Y(y) &= C\sin(ky) + D\cos(ky) \end{aligned}$$

And thus:

$$V(x,y) = (Ae^{kx} + Be^{-kx})(C\sin(ky) + D\cos(ky))$$

We then use our BCs to find the coefficients:

$$\begin{aligned} V \to 0 \text{ as } x \to \infty &\to A = 0 \\ V(x,0) = 0 &\to D = 0 \\ V(x,y) = 0 &\to k = \frac{n\pi}{a}, (n = 1, 2, 3, \dots) \end{aligned}$$

Since any linear combination of solutions the Laplace equation is also a solution, we can construct a solution by taking the infinite sum:

$$V(x,y) = \sum_{n=1}^{\infty} C_n e^{-\frac{n\pi x}{a}} \sin\Big(\frac{n\pi y}{a}\Big)$$

Using our initial conditions we get the **Fourier sine series**:

$$V(0,y) = \sum_{n=1}^{\infty} C_n \sin\Big(\frac{n\pi y}{a}\Big) = V_0(y)$$

We can find the coefficients C_n by multiplying both sides by $\sin\big(\frac{m\pi y}{a}\big)$ and integrating, which is called **Fourier's trick**, which (after rearrangement) gives us:

$$C_n = \frac{2}{a}\int_0^a V_0(y)\sin\Big(\frac{n\pi y}{a}\Big)dy$$

⊛

We can use similar methods to solve for V in other situations with given boundary conditions, and in general we take the following steps:

1. Separate V into functions of the relevant coordinates
2. Write the Laplace equation using the product functions and define the eigenvalues (Note: In three dimensions, an eigenvalue might be the sum of the squares of two others)
3. Use the boundary conditions to solve for coefficients
4. Use Fourier's trick to find the remaining coefficient (Note: In three dimensions, may have product of two harmonic functions)

3.3.2 Spherical Coordinates

In spherical coordinates, if a problem has azimuthal symmetry (meaning that V is independent of ϕ), then the Laplace equation is:

$$\frac{\partial}{\partial r}(r^2\frac{\partial V}{\partial r}) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial V}{\partial\theta}) = 0$$

We assume that the solution is a product of functions in r and θ:

$$V(r,\theta) = R(r)\Theta(\theta)$$

Like before, we plug this back into the Laplace equation and determine that each term is a constant (which we call $l(l+1)$ to make our math easier later), eventually giving us:

$$\frac{d}{dr}(r^2\frac{dR}{dr}) = l(l+1)R$$
$$\frac{d}{d\theta}(\sin\theta\frac{d\Theta}{d\theta}) = -l(l+1)\sin\theta\Theta$$

The first ODE is a Cauchy-Euler equation, so we easily determine that its solution is:

$$R(r) = Ar^l + \frac{B}{r^{l+1}}$$

The second ODE turns out to have solutions that are **Legendre polynomials** in the variable $\cos\theta$:

$$\Theta(\theta) = P_l(\cos\theta)$$

$P_l(x)$ is conveniently defined by the **Rodrigues formula**:

$$P_l(x) = \frac{1}{2^l l!}(\frac{d}{dx})^l(x^2-1)^l$$

Note that $P_l(x)$ is an lth-order polynomial in x and contains only even powers of x if l is even and only odd powers if l is odd.

While the ODE of Θ is second order, it turns out that the non-Legendre polynomial solutions blow up at $\theta = 0$ or $\theta = \pi$ and are thus physically impossible, meaning we can ignore them. Therefore, the general solution is the linear combination of solutions:

$$V(r,\theta) = \sum_{l=0}^{\infty}(A_l r^l + \frac{B_l}{r^{l+1}})P_l(\cos\theta)$$

Like in Cartesian coordinates, the general method is to write out the product solutions and then use the boundary conditions (and other known information) to solve for the coefficients. Fourier's trick is of use again, and for Legendre polynomials is:

$$\int_0^\pi P_l(\cos\theta)P_{l'}(\cos\theta)\sin\theta d\theta = \begin{cases} 0 & l \neq l' \\ \frac{2}{2l+1} & l = l' \end{cases}$$

3.4 Method of Relaxation

The **method of relaxation** essentially uses numerical differentiation to find an approximate solution to the Laplace equation.

In two dimensions, the partial derivatives of the potential are:

$$\begin{aligned}
\frac{\partial V}{\partial x} &\approx \frac{V(x+\frac{\epsilon}{2},y)-V(x-\frac{\epsilon}{2},y)}{\epsilon} \\
\frac{\partial V}{\partial y} &\approx \frac{V(x,y+\frac{\epsilon}{2})-V(x,y-\frac{\epsilon}{2})}{\epsilon} \\
\frac{\partial^2 V}{\partial x^2} &\approx \frac{V(x+\epsilon,y)-2V(x,y)+V(x-\epsilon),y}{\epsilon^2} \\
\frac{\partial^2 V}{\partial y^2} &\approx \frac{V(x,y+\epsilon)-2V(x,y)+V(x,y-\epsilon)}{\epsilon^2}
\end{aligned}$$

Thus, the Laplace equation becomes:

$$\frac{\partial^2 V}{\partial x^2}+\frac{\partial^2 V}{\partial y^2} \approx \frac{1}{\epsilon^2}[V(x+\epsilon,y)+V(x-\epsilon,y)+V(x,y+\epsilon)+V(x,y-\epsilon)-4V(x,y)]=0$$

We solve for $V(x,y)$:

$$V(x,y) \approx \frac{1}{4}[V(x+\epsilon,y)+V(x-\epsilon,y)+V(x,y+\epsilon)+V(x,y-\epsilon)]$$

Therefore, on a discrete grid with spacing ϵ, V at a point is the average of its values at the four nearest points. We can thus approximate the solution computationally by assigning values to points on the edge using the boundary conditions and then sequentially finding the value for interior points. This is most commonly done by assigning an initial value (like 0) to the interior points, rastering through all of them, and then repeatedly averaging until the desired precision is reached.

3.5 Multipole Expansion

Suppose we have a region density $\rho(\mathbf{r}')$. Then, the potential outside this region is:

$$V(\mathbf{r})=\frac{1}{4\pi\epsilon_0}\int \frac{1}{\boldsymbol{\imath}}\rho(\mathbf{r}')d\tau'$$

If we take some point within the region and let α be the angle between $\mathbf{r}$ and $\mathbf{r}'$, then the law of cosines gives us:

$$\boldsymbol{\imath}^2 = r^2+r'^2-2rr'\cos\alpha = r^2[1+(\frac{r'}{r})^2-2(\frac{r'}{r})\cos\alpha]$$

Thus, we can rewrite the separation vector as:

$$\boldsymbol{\imath} = r\sqrt{1+\epsilon}, \epsilon = (\frac{r'}{r})(\frac{r'}{r}-2\cos\alpha)$$

At points far outside of the charge distribution, ϵ is very small, so we can use the binomial expansion to find $\frac{1}{\boldsymbol{\imath}}$:

$$\frac{1}{\boldsymbol{\imath}}=\frac{1}{r}(1+\epsilon)^{-\frac{1}{2}}=\frac{1}{r}(1-\frac{1}{2}\epsilon+\frac{3}{8}\epsilon^2-\frac{5}{16}\epsilon^3+\dots)$$

Substituting in ϵ, we realize that this is just a sum of Legendre polynomials:

$$\begin{aligned}
\frac{1}{r} &= \frac{1}{r}[1-\frac{1}{2}(\frac{r'}{r})(\frac{r'}{r}-2\cos\alpha)+\frac{3}{8}(\frac{r'}{r})^2(\frac{r'}{r}-2\cos\alpha)^2+\dots] \\
&= \frac{1}{r}[1+\frac{r'}{r}P_1(\cos\alpha)+(\frac{r'}{r})^2P_2(\cos\alpha)+\dots] \\
&= \frac{1}{r}\sum_{n=0}^{\infty}(\frac{r'}{r})^nP_n(\cos\alpha)
\end{aligned}$$

Thus, the potential is:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\sum_{n=0}^{\infty}\frac{1}{r^{n+1}}\int r'^n P_n(\cos\alpha)\rho(\mathbf{r}')d\tau'$$

This is called the **multipole expansion** of the potential. The first term is the **monopole** term, the second is the **dipole**, the third is the **quadrupole**, the fourth is the **octopole**, and so on. Each successive term is a factor of r less than the previous term, so at large distances, we can approximate the potential with the earliest non-vanishing term.

3.5.1 Dipoles

The dipole term can be written more succinctly as:

$$V_{dip}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}$$

where $\mathbf{p}$ is the **dipole moment** of the charge distribution:

$$\mathbf{p} \equiv \int \mathbf{r}'\rho(\mathbf{r}')d\tau'$$

A **physical dipole** is a configuration of two equal and opposite point charges with separation vector $\mathbf{d}$. The dipole moment of a physical dipole is thus:

$$\mathbf{p} = q\mathbf{r}'_+ - q\mathbf{r}'_- = q(\mathbf{r}'_+ - \mathbf{r}'_-) = q\mathbf{d}$$

A **perfect dipole** is something that only has a dipole moment, with no quadrupole or higher terms. This is the limit when $d \to 0$ and $q \to \infty$ if the product is constant.

> **Note.** The monopole moment does not depend on the origin, but dipole and higher terms ordinarily do. However, if the total charge is zero, the dipole moment doesn't depend on the choice of origin.

3.5.2 The Electric Field of a Dipole

The electric field of a perfect dipole is:

$$\mathbf{E}_{dip}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}[3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]$$

In spherical coordinates, this is:

$$\mathbf{E}_{dip}(r,\theta) = \frac{p}{4\pi\epsilon_0 r^3}(2\cos\theta\hat{\mathbf{r}} + \sin\theta\hat{\theta})$$

4 Electric Fields in Matter

4.1 Polarization

Definition 4.1. A **dielectric** is an insulator that can be **polarized** by an external electric field, with charges shifting or rotating slightly.

If an electric field is applied to an atom, the centers of the electron cloud and nucleus shift and no longer coincide, forming an induced dipole moment in the same direction as the external field:

$$\mathbf{p} = \alpha \mathbf{E}$$

where α is the **atomic polarizability**. For simple atoms, this is approximately proportional to atomic volume.

Molecules often polarize more easily in certain directions than others, so we replace α with the **polarizability tensor** with nine components α_{ij}, giving us the following equations:

$$\begin{aligned} p_x &= \alpha_{xx}E_x + \alpha_{xy}E_y + \alpha_{xz}E_z \\ p_y &= \alpha_{yx}E_x + \alpha_{yy}E_y + \alpha_{yz}E_z \\ p_z &= \alpha_{zx}E_x + \alpha_{zy}E_y + \alpha_{zz}E_z \end{aligned}$$

Polar molecules are molecules that have dipole moments. If a dipole is placed in a uniform electric field, it experiences a net force of zero, but a nonzero torque that causes it to rotate until it is parallel to the field:

$$\mathbf{N} = \mathbf{p} \times \mathbf{E}$$

If a dipole is placed in a nonuniform electric field, the force it experiences is nonzero:

$$\mathbf{F} = q(\mathbf{E}_+ - \mathbf{E}_-) = q(\Delta \mathbf{E}) = (\nabla E_x) \cdot \mathbf{d} = (\mathbf{d} \cdot \nabla)\mathbf{E} = (\mathbf{p} \cdot \nabla)\mathbf{E}$$

The energy of a permanent ideal dipole in an electric field is:

$$U = -\mathbf{p} \cdot \mathbf{E}$$